EEE231 DIGITAL ELECTRONICS LABORATORY

SESSION 2025/2026

MINI PROJECT REPORT

Title: Photon War

GROUP 19

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1.0 Theme and Title

1.1 Theme

The theme of the EEE231 Digital Electronics Laboratory mini project is “Sci-fi themed game”. We are required to build a game with the theme of Sci-fi by applying the knowledge and theory of digital circuits.

1.2 Title

“Photon War” was chosen as the title of this mini project.

2.0 Aim And Objective

2.1 Aim

To construct a mini-game, "Photon War," using electronic components that demonstrate digital logic theories and circuit-building skills.

2.2 Objectives

- To design interactive LED lane circuits where two players can press their corresponding button to shift the Photon (active LED of the lane) toward their opponent's side.
- To design a circuit that blocks the player input when the Photon reaches one end of the lane, and determine the winner of every lane.
- To design a score counter circuit to calculate the total score of the player in the three lanes, and determine the winner of the game.

3.0 Introduction

“Photon War” is a competitive two-player game where players press buttons assigned to specific lanes to push “Photon” toward the opponent's side. Once a Photon reaches the end of a lane, that lane is conquered. Victory is determined by the total number of lanes successfully conquered.

The game is based on the concept of deep space warfare, where opposing forces struggle to dominate galactic sectors. Players assume the role of strategic commanders, directing their respective forces to conquer and secure dominance over disputed galactic sectors. The game’s mechanics are designed to be intuitive yet engaging, making the immersive sci-fi role-play accessible to a broad demographic, including children, adults, and seniors.

To start the game, a reset button must be pressed to ensure all Photons are returned to their initial positions at the center of each lane. The game features three concurrent lanes, each consisting of 15 RGB LEDs. The interface includes three red buttons for Player Red and three blue buttons for Player Blue. While players may freely choose which lane to engage, they are restricted to using only a single finger throughout the duration of the match. A green light indicates the Photon in every lane, shifting incrementally toward the opponent’s side each time a player presses their corresponding button.

When a player manages to move the Photon to the opponent’s end, the entire lane lights up with the winner’s color: red for Player Red and blue for Player Blue. Once all three lanes are completed, the letter “W” will be shown on the 16-segment display at the winner's side to indicate the winner of the game.

3.1 Problem Statement

The problem that we encountered in the design and development of this mini project are as follow:

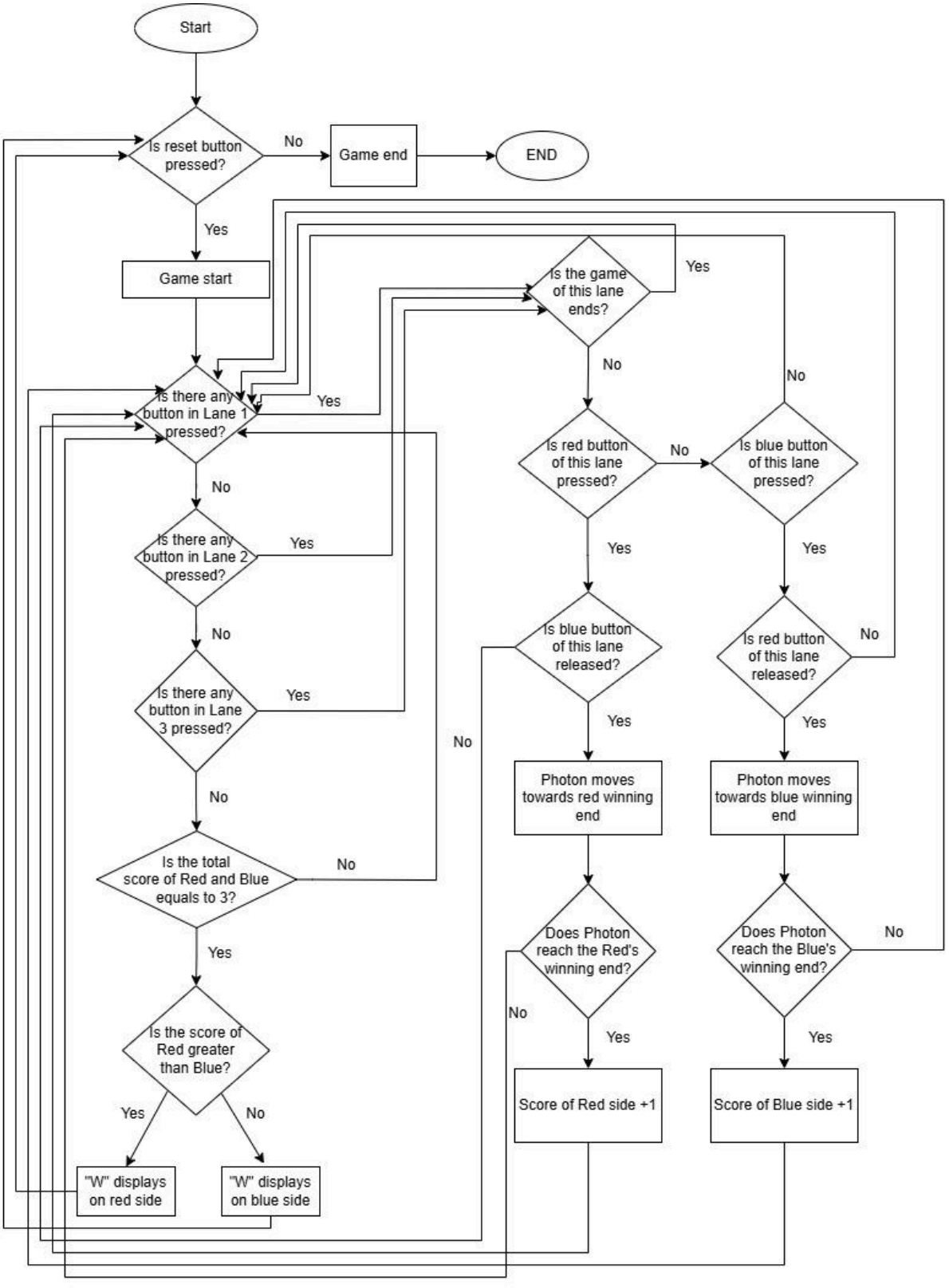
1. How to visualise the movement of the Photon and the “conquer” effect (the entire lane lighting up in the winner’s color)?
2. How to enable two separate button inputs to control the bidirectional movement of a Photon across 15 LEDs?
3. How to resolve the "button bounce" issue to ensure precise input detection?
4. How to prevent circuit glitches when both buttons in the same lane are pressed simultaneously?
5. How to disable button inputs once a lane has been conquered?
6. How to accurately determine the overall winner of the game?
7. How to allow the “W” displays on the winner’s side only after all three lanes have finished?
8. How to ensure all Photons are reset to the center of their respective lanes before the game begins?

3.2 Proposed Solution

1. **LED Configuration:** 15 RGB LEDs are utilized per lane; red and blue are reserved for the "conquer" effect, while green indicates Photon movement.
2. **Movement Control:** Up/Down BCD Counters receive the two button inputs to facilitate back-and-forth movement. A 4-to-16 decoder (constructed from two 3-to-8 decoders) is used to translate the BCD output and trigger the corresponding LEDs.
3. **Debounce Circuitry:** A button debounce circuit featuring an RC filter and a Schmitt Trigger is implemented to clean the input signals.
4. **Input Protection:** An SR latch circuit is used to block button inputs if the opponent's button in the same lane has not yet been released.
5. **Input Termination:** Combinational logic is used to inhibit further input signals once the circuit detects that a Photon has reached the end of a lane, preventing movement after a victory.
6. **Logic Processing:** Adders and comparators are used to tally the lanes conquered by each player and determine the final winner.
7. **Display Logic:** Combinational circuits detect the completion status of all lanes, enabling the 16-segment display the character "W" on the winner's side only when the game concludes.
8. **System Reset:** A counter IC with a customizable preset feature is used to load the initial starting sequence into the counter before the game begins.

4.0 Methodology

4.1 Flow Chart



4.2 Photon War Circuit

The Photon War circuit consist of 2 main part:

- LED Lane Circuit
- Winner Indicator Circuit

4.2.1 LED Lane Circuit

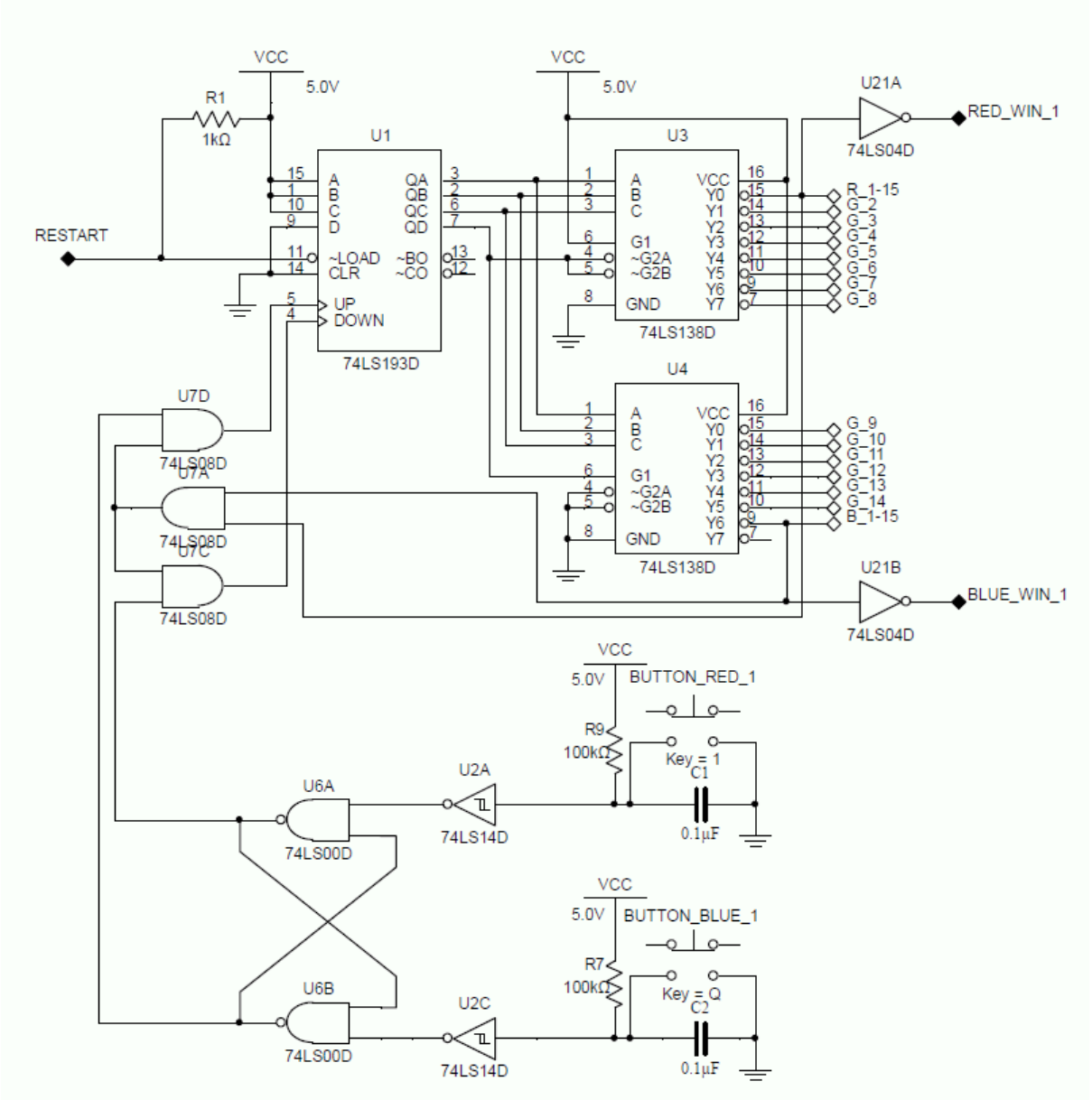


Figure 4.2.1(a)

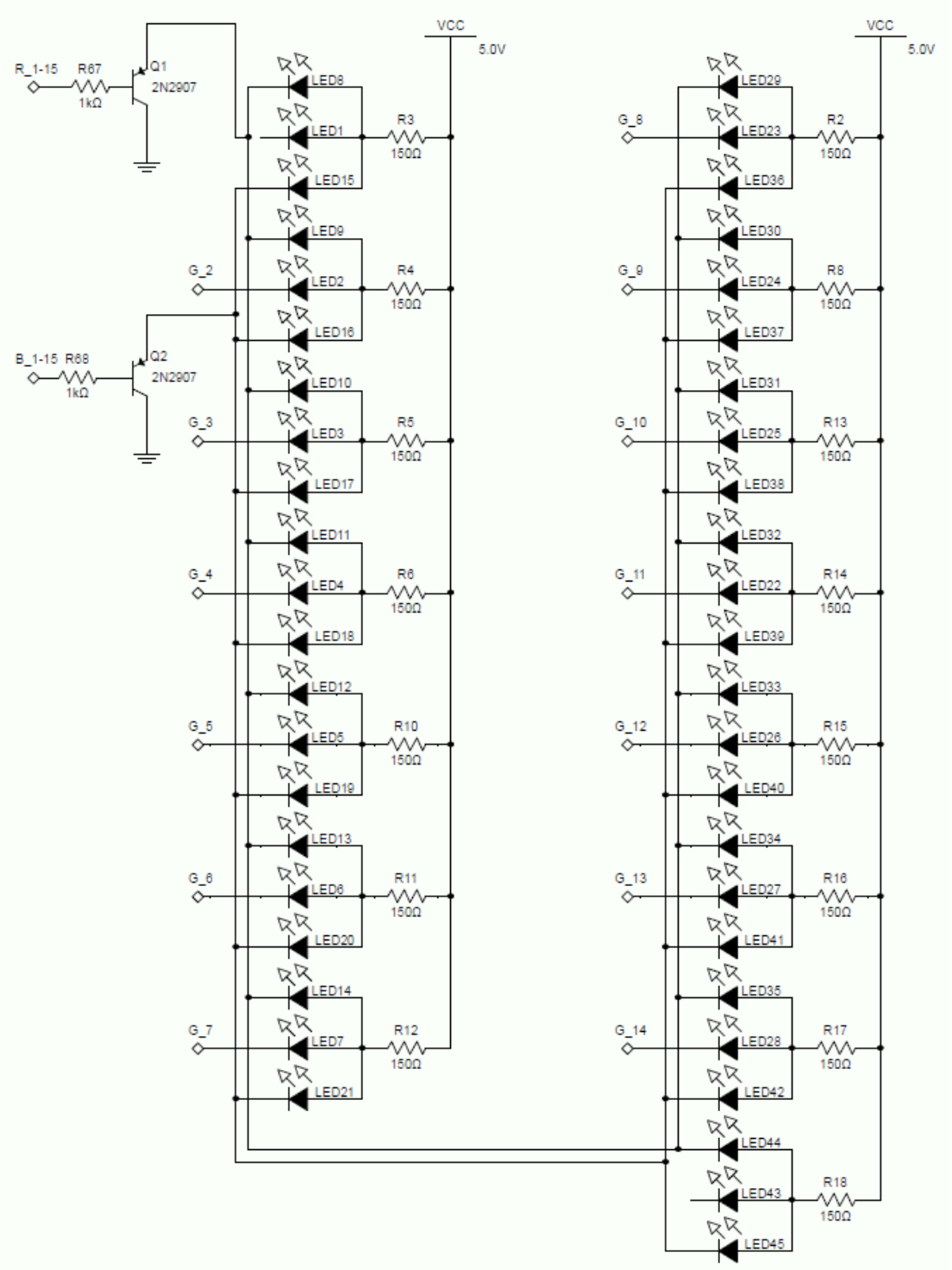


Figure 4.2.1(b)

Figure 4.2.1(a) and (b) show the complete LED lane circuit for a single lane. The circuit consists of 4 parts: Photon Movement Control Circuit, LED Array Circuit, Debounced Circuit, and Glitch Prevention Circuit.

4.2.1.1 Movement Control Circuit

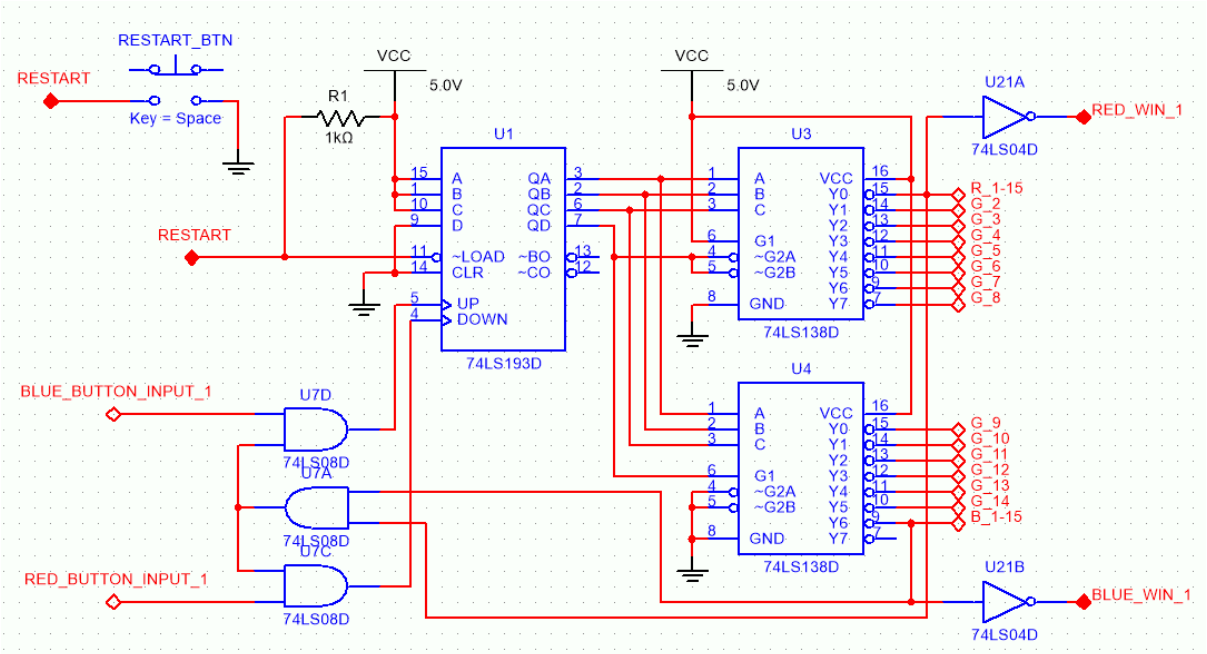


Figure 4.2.1.1

Figure 4.2.1.1 shows the Photon Movement Control Circuit. The “Photon” is defined as the singular active green LED that tracks the counter's output across the RGB LED Array. 74LS193 (Up/Down BCD Counter) and two 74LS138 (3-to-8 decoder) are the main ICs to control the movement of the Photon.

74LS193 features two separate clock inputs for incrementing and decrementing the count, making it ideal for the movement of the Photon. 74LS193 has two clock signal pins, UP and DOWN. UP is connected to the pulses generated by the Player Blue presses and DOWN is connected to HIGH to allow the 74LS193 to count incrementally while DOWN is connected to the pulses generated by the Player Red presses and UP is connected to HIGH to allow the 74LS193 to count decrementally. 74LS193 has 4-bit binary output: Q_D (MSB), Q_C, Q_B, and Q_A (LSB), representing 16 discrete states, 0000_2 (0_{10}) to 1111_2 (15_{10}), which are sent to the decoders to determine the photon's position. The LED array requires an odd number of LEDs to allow the initial position of Photon to be in the middle of the lane. Thus, the output states 0_{10} to 14_{10} is used and 15_{10} is unused. The active low parallel load pin, $\overline{LOAD}$ allows the circuit to be preset to a specific starting value at the beginning of each round that can be configured via PRESET pins: D (MSB), C, B, and A (LSB), which in this circuit is configured to 0111_2 (7_{10}). The $\overline{LOAD}$ pin is connected to a pull-up resistor connected to 5V,

with the button providing a path to GND. When the button is pressed, the counter is loaded with the preset value, 0111_2.

Two 74LS138 form a 4-to-16 decoder, which consist of 16 outputs and all outputs are used to drive the RGB LEDs to visualise the movement of Photon and the last output of the 4-to-16 output is unused. 74LS138 consists of one active high enable pin, G1, and two active low enable pins, $\overline{G2A}$ and $\overline{G2B}$. All enable pins must be activated to allow 74LS138 to work. To cascade two 74LS138 into a 4-to-16 decoder, G1 is connected to HIGH and $\overline{G2A}$ and $\overline{G2B}$ is connected to Q_D from 74LS193 for the first 74LS138 (outputs 0–7) whereas G1 is connected to Q_D from 74LS193, $\overline{G2A}$ and $\overline{G2B}$ is connected to LOW for the second 74LS138 (outputs 8–14). Both 74LS138 shares the Q_C, Q_B, and Q_A from the 74LS193, which are connected to their respective C, B, and A input pins.

Three 74LS08 AND gates are used to inhibit further input signals once the circuit detects that a Photon has reached the end of a lane, preventing movement after a victory of the lane. The first AND gate receives inputs from the first output of 4-to-16 decoder ($\overline{Y_0}$ pin of the first 74LS138) and the 15th output of the 4-to-16 decoder ($\overline{Y_6}$ pin of the second 74LS138). The pulses generated by the Blue and Red player buttons are fed into two subsequent AND gates, which also receive the output from the first AND gate before connecting to the UP and DOWN pins of the 74LS193. When the Photon is in the field of play, the first AND gate receives two HIGH signals and outputs a HIGH state, allowing the button pulses to pass through to the counter. However, once the Photon reaches either end of the LED array, the first AND gate receives a LOW signal from the active-low decoder, forcing its output LOW and successfully blocking any further pulses from reaching the 74LS193.

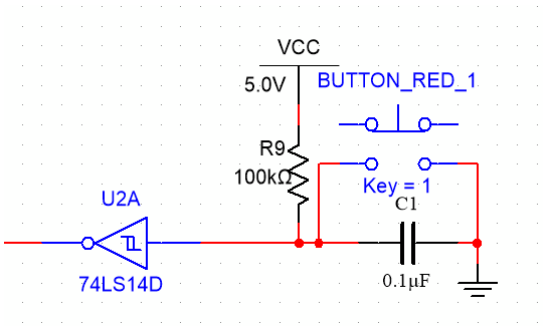
Two 74LS04 inverters are used to invert the first active low output ($\overline{Y_0}$ pin of the first 74LS138) and the 15th output of the 4-to-16 decoder ($\overline{Y_6}$ pin of the second 74LS138) for the Score Counter and Winner Indicator Circuit.

4.2.1.2 LED Array Circuit

As shown in Figure 4.2.1(b), the LED Array part consists of 15 common anode RGB LEDs, 15 150Ω resistors and 2 PNP BJTs. Common anode pins of 15 RGB LEDs are connected to 5V with 150Ω resistors. Usually RGB LEDs require 3 resistors per RGB LED but since every RGB LEDs only lit up one colour at once, it is safe to let the three colours of a single RGB LED share the same resistor to save the number of resistors used.

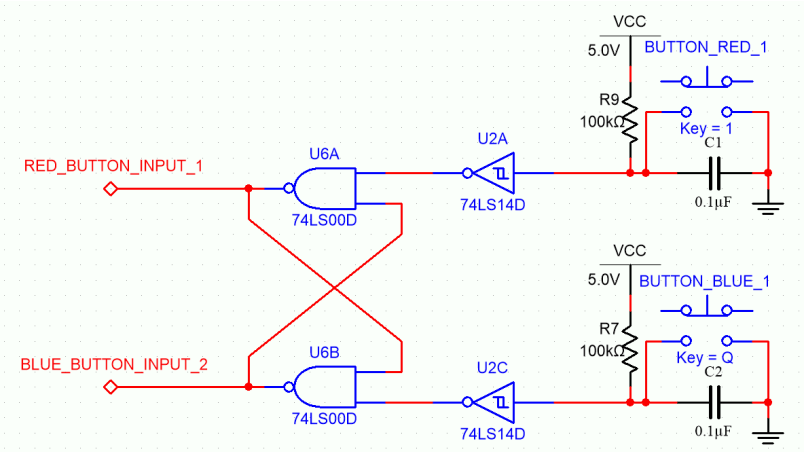
All red cathode pins of 15 RGB LEDs are connected to a PNP BJT because the 74LS138 cannot sink enough current. Similarly, the blue cathode pins are also driven by their own PNP BJT. The base of the PNP BJT that drives the red cathode pins are connected to the first output of the 4-to-16 decoder, base of the PNP BJT that drives the blue cathode pins are connected to the 15th output, whereas the green cathode pins of the second to 14th RGB LEDs are connected to the second to 14th output of the 4-to-16 decoder. The green cathode pin of the first and the 15th RGB LEDs are not used. The 16th output of the 4-to-16 decoder is also not used.

4.2.1.3 Debounced Circuit



To ensure reliable state transitions, the player input buttons are conditioned using a passive RC low-pass filter followed by a 74LS14 Schmitt Trigger inverter. Mechanical tactile switches inherently suffer from "contact bounce," where the physical metal plates vibrate and create several rapid logic transitions during a single press. An RC network with a 10ms time constant was used to smooth the rapid voltage spikes into a gradual analog curve. This signal is then fed into the Schmitt Trigger, which utilizes hysteresis to convert the slow-moving voltage into a sharp, clean digital pulse. This multi-stage debouncing process is critical for the 74LS193 counter, as it prevents a single button press from being registered as multiple "Photon" movements.

4.2.1.4 Glitch Prevention Circuit



The 74LS193 may exhibit unstable behavior or fail to register the transition when one clock input receives pulses while the other clock input is held LOW. To prevent logic glitches within the 74LS193, an SR latch from cross-coupled 74LS00 NAND gates, is implemented as a signal arbitrator. The circuit ensures a "mutually exclusive" logic state where both clock outputs default to HIGH when no buttons are pressed. The latch's inherent memory function handles simultaneous button presses by maintaining the current state to prevent unstable behaviour of the 74LS193. This ensures that only one valid rising edge is delivered to either the UP or DOWN pin and the other pin is held HIGH at any given time, maintaining the integrity of the Photon's movement.

4.2.2 Winner Indicator Circuit

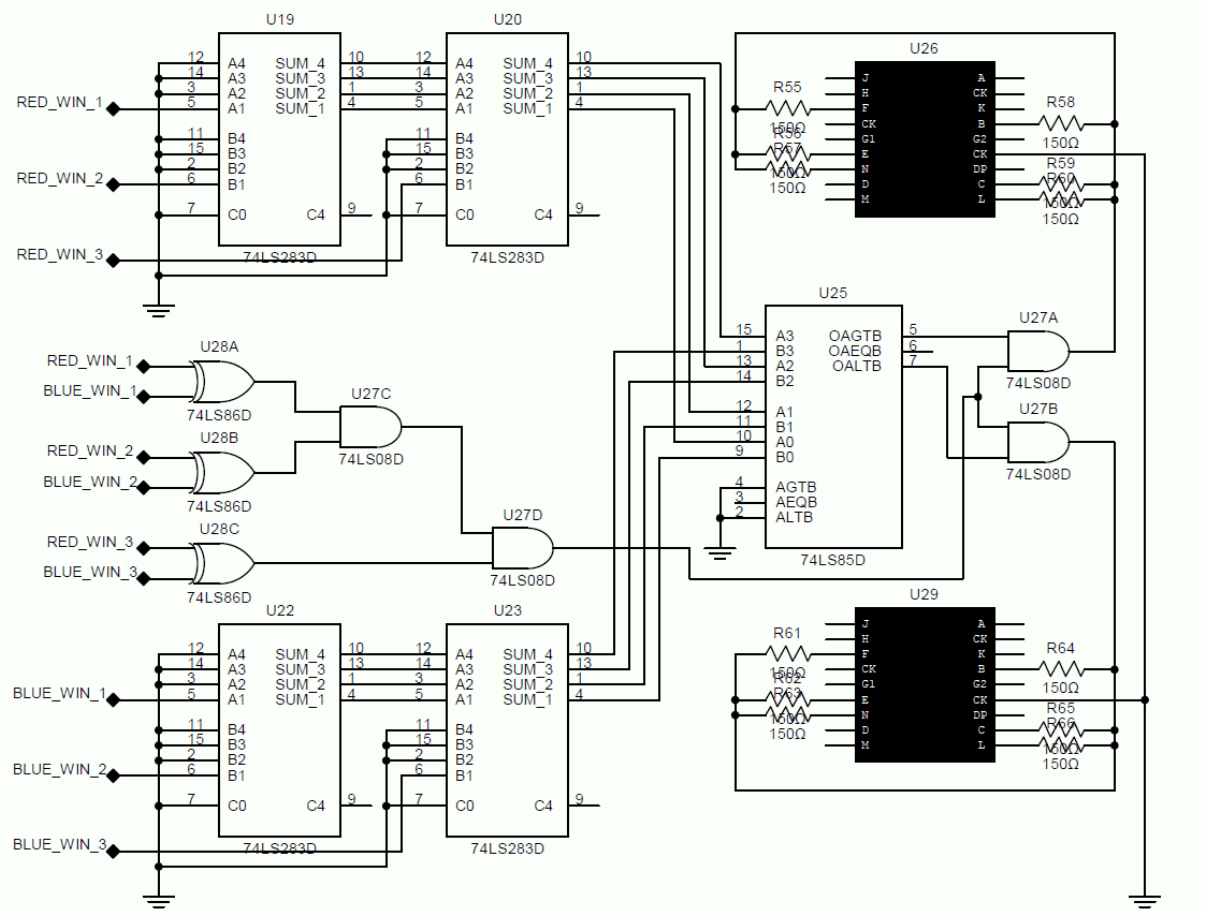


Figure 4.2.2

Figure 4.2.2 shows the Winner Indicator Circuit. 74LS283 Adder is used to obtain the score of Player Red and Blue by receiving the inverted first and the 15th output of the 4-to-16 decoder of the three lanes and adding up the number of winning lanes of the player Red and Blue. Then, the output of the Player Red and Blue score is fed to the 74LS85 Comparator to determine the player with the highest score.

The 74LS86 XOR gate is used to track whether the single lane has a winner by receiving the inverted first and the 15th output of the 4-to-16 decoder. XOR gates outputs LOW at default as it receives two LOW inputs during the game. When the Photon reaches either end of the LED array, XOR gates receive one HIGH input thus outputs HIGH. The XOR gates of all three lanes are then connected to a 3-input AND gate (forms by cascading two 2-input 74LS08 AND gates) and only outputs HIGH when every lane has a winner. The output of the comparator is fed into two subsequent AND gates, which also receive the output

from the 3-input AND gate before connecting to the 16-segment display of each player. The 16-segment display will display the “W” letter and at the winner’s side when the all three lanes are ended to indicate the final winner of the game.

5.0 Result and Discussion

5.1 Result of Photon War Circuit

5.1.1 Debounced Circuit

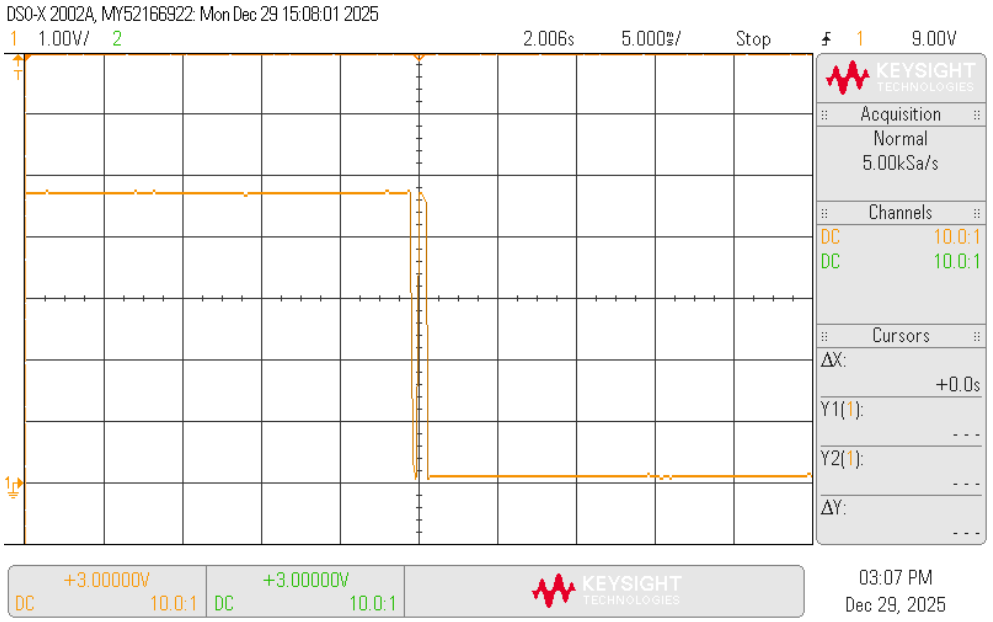


Figure 5.1.1 (a)

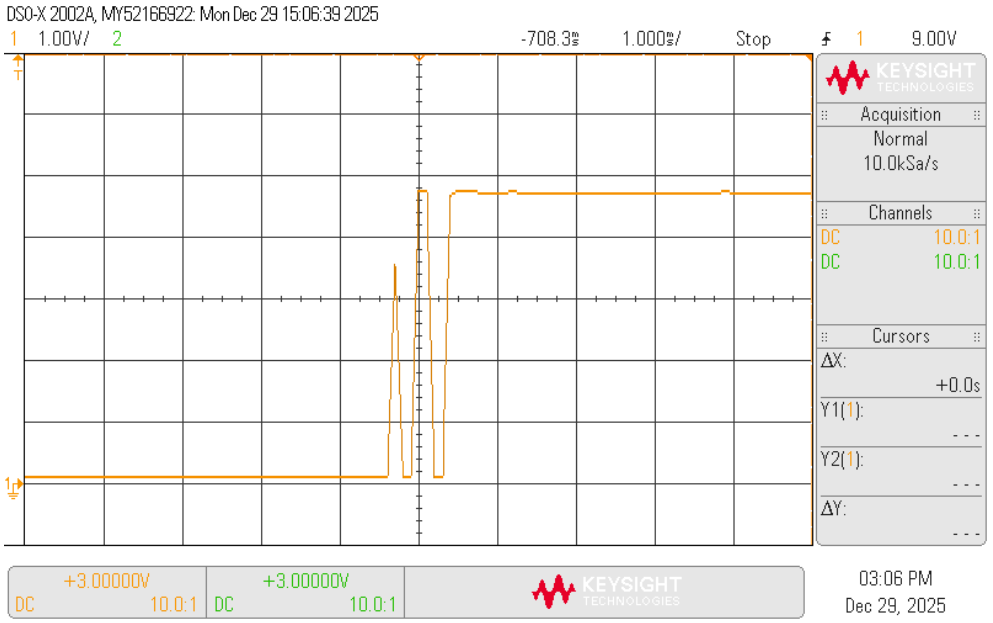


Figure 5.1.1 (b)

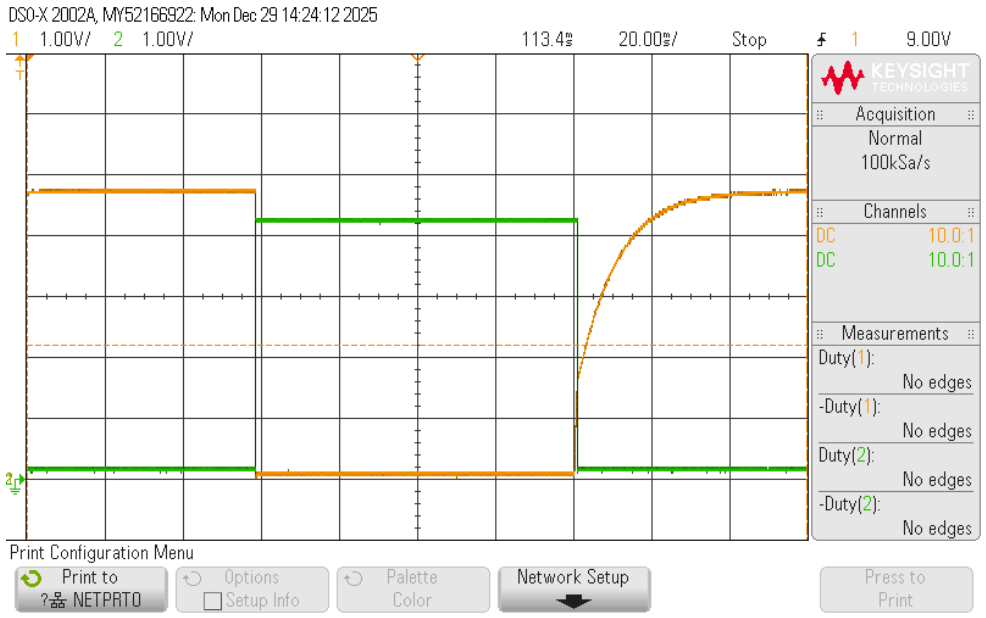


Figure 5.1.1 (c)

Figures 5.1.1 (a) and (b) shows the contact bounce waveforms of the tactile button when pressed and released. Figure 5.1.1 (c) displays the debounced waveforms: the yellow trace represents the output using only the RC filter, while the green trace shows the final output after passing through both the RC filter and the Schmitt Inverter. As observed in Figure 5.1.1 (c), the RC filter (yellow waveform) successfully eliminates the high frequency contact bounce. However, it introduces a slow rise time resulting in an exponential ramp that lingers near the switching boundary of a standard logic gate, any noise can cause the input to cross the threshold multiple times, leading to multiple triggering at the output. By passing this signal through a Schmitt Inverter (green waveform), the circuit utilizes hysteresis to convert the slow-moving ramp into a sharp, clean digital pulse. This ensures the subsequent logic gates receive a stable transition without multiple triggering.

5.1.2 Movement Control Circuit

Decimal State	Counter Binary Output ($Q_DQ_CQ_BQ_A$)	Active Decoder	Active Pin	LED Colour & Position
0	0000	Decoder 1	$\sim Y_0$	Red 1-15 (Goal - Red Win)
1	0001	Decoder 1	$\sim Y_1$	Green 2
2	0010	Decoder 1	$\sim Y_2$	Green 3
3	0011	Decoder 1	$\sim Y_3$	Green 4
4	0100	Decoder 1	$\sim Y_4$	Green 5
5	0101	Decoder 1	$\sim Y_5$	Green 6
6	0110	Decoder 1	$\sim Y_6$	Green 7
7	0111	Decoder 1	$\sim Y_7$	Green 8 (START)
8	1000	Decoder 2	$\sim Y_0$	Green 9
9	1001	Decoder 2	$\sim Y_1$	Green 10
10	1010	Decoder 2	$\sim Y_2$	Green 11
11	1011	Decoder 2	$\sim Y_3$	Green 12
12	1100	Decoder 2	$\sim Y_4$	Green 13
13	1101	Decoder 2	$\sim Y_5$	Green 14
14	1110	Decoder 2	$\sim Y_6$	Blue 1-15 (Goal - Blue Win)
15	1111	N/A	N/A	Inhibited

Table 5.1.2 (a)

Table 5.1.2 (a) shows the complete state sequence of the binary output and the decoder output. This shows that at State 7 (0111_2), the circuit is using Decoder 1, but at State 8 (1000_2), it successfully switches to Decoder 2. The transition from State 7 to State 8 demonstrates the successful toggle of the enable pins via Q_D, effectively cascading the two 3-to-8 decoders into a single 4-to-16 decoder. All red parts of the RGB LEDs light up at State 0 and all blue parts of the RGB LEDs light up at State 14. State 15 is unused.

Button in Action	Counter Binary Output ($Q_DQ_CQ_BQ_A$)	Active Decoder Output	Result
RESTART	0111	~Y7 (Dec 1)	LOAD pin pulls LOW; presets to 7. Photon starts in the center position.
BLUE	1000	~Y0 (Dec 2)	Counter increments; Photon moves.
RED	0111	~Y7 (Dec 1)	Counter decrements; Photon moves.
BLUE	1000	~Y0 (Dec 2)	Counter increments; Photon moves.
RED	0111	~Y7 (Dec 1)	Counter decrements; Photon moves.
RED	0110	~Y6 (Dec 1)	Counter decrements; Photon moves.
RED	0101	~Y5 (Dec 1)	Counter decrements; Photon moves.
BLUE	0110	~Y6 (Dec 1)	Counter increments; Photon moves.
RED	0101	~Y5 (Dec 1)	Counter decrements; Photon moves.
RED	0100	~Y4 (Dec 1)	Counter decrements; Photon moves.
RED	0011	~Y3 (Dec 1)	Counter decrements; Photon moves.
RED	0010	~Y2 (Dec 1)	Counter decrements; Photon moves.
BLUE	0011	~Y3 (Dec 1)	Counter increments; Photon moves.
BLUE	0100	~Y4 (Dec 1)	Counter increments; Photon moves.
RED	0011	~Y3 (Dec 1)	Counter decrements; Photon moves.
RED	0010	~Y2 (Dec 1)	Counter decrements; Photon moves.
RED	0001	~Y1 (Dec 1)	Counter decrements; Photon moves.
RED	0000	~Y0 (Dec 1)	Counter increments; Player Red wins the lane.
RED	0000	~Y0 (Dec 1)	AND gates block pulse; counter locks.
BLUE	0000	~Y0 (Dec 1)	AND gates block pulse; counter locks.
RESTART	0111	~Y7 (Dec 1)	LOAD pin pulls LOW; presets to 7. Photon starts in the center position.

Table 5.1.2 (b)

Button in Action	Counter Binary Output ($Q_DQ_CQ_BQ_A$)	Active Decoder Output	Result
RESTART	0111	~Y7 (Dec 1)	LOAD pin pulls LOW; presets to 7. Photon starts in the center position.
BLUE	1000	~Y0 (Dec 2)	Counter increments; Photon moves.
RED	0111	~Y7 (Dec 1)	Counter decrements; Photon moves.
BLUE	1000	~Y0 (Dec 2)	Counter increments; Photon moves.
RED	0111	~Y7 (Dec 1)	Counter decrements; Photon moves.
BLUE	1000	~Y0 (Dec 2)	Counter increments; Photon moves.
BLUE	1001	~Y1 (Dec 2)	Counter increments; Photon moves.
BLUE	1010	~Y2 (Dec 2)	Counter increments; Photon moves.
BLUE	1011	~Y3 (Dec 2)	Counter increments; Photon moves.
RED	1010	~Y2 (Dec 2)	Counter decrements; Photon moves.
BLUE	1011	~Y3 (Dec 2)	Counter increments; Photon moves.
BLUE	1100	~Y4 (Dec 2)	Counter increments; Photon moves.
BLUE	1101	~Y5 (Dec 2)	Counter increments; Photon moves.
RED	1100	~Y4 (Dec 2)	Counter decrements; Photon moves.
RED	1011	~Y3 (Dec 2)	Counter decrements; Photon moves.
BLUE	1100	~Y4 (Dec 2)	Counter increments; Photon moves.
BLUE	1101	~Y5 (Dec 2)	Counter increments; Photon moves.
BLUE	1110	~Y6 (Dec 2)	Counter increments; Player Blue wins the lane.
BLUE	1110	~Y6 (Dec 2)	AND gates block pulse; counter locks.
RED	1110	~Y6 (Dec 2)	AND gates block pulse; counter locks.
RESTART	0111	~Y7 (Dec 1)	LOAD pin pulls LOW; presets to 7. Photon starts in the center position.

Table 5.1.2 (c)

Table 5.1.2 (b) and (c) shows the movement of Photon when Player Red or Blue pressed their button. When player Red or Blue press their button, the counter counts decrementally or incrementally and the Photon moves back or forth. When the Photon reaches one end of the LED Array, all AND gates output in that lane is forced LOW regardless of the button input, thus blocking the player button input pulse to the counter and the further movement of the Photon.

5.1.3 Glitch Prevention Circuit

Button Input (after Debounced Circuit)		Button Input (after Glitch Prevention)		NAND gate SR-Latch State	Result
BLUE (S)	RED (R)	UP_{CLOCK} (Q')	$DOWN_{CLOCK}$ (Q)		
0 (Idle)	0 (Idle)	1	1	Invalid	-
0 (Idle)	1 (Pressed)	1	0	Reset	Red is pressed.
1	1	1	0	No Change	Red is not released yet. Blue Input is blocked
0	1	1	0	Reset	-
1	1	1	0	No Change	Red is not released yet. Blue Input is blocked
0	1	1	0	Reset	-
0	0	1	1	Invalid	Red is released. Counter decrements; Photon moves.
1	0	0	1	Set	Blue is pressed.
1	1	0	1	No Change	Blue is not released yet. Red Input is blocked
0	1	1	0	Reset	Blue is released. Counter increments; Photon moves.
0	0	1	1	Invalid	Red is released. Counter decrements; Photon moves.

Table 5.1.3

Table 5.1.3 shows the real-time behavior of the Glitch Prevention Circuit by comparing the debounced button inputs to the resulting clock signals delivered to the 74LS193. When no buttons are pressed, the SR latch is maintained in an “Invalid” state (where both inputs are LOW), which effectively forces both outputs to remain HIGH. As a player presses and subsequently releases a button, the latch transitions from this “Invalid”

state to either a “Set” or “Reset” state before returning to the default HIGH state. By utilizing the latch's “Invalid” state for the button's idle condition, the circuit ensures that every button release automatically generates the required rising edge to HIGH. Furthermore, when both players press their buttons simultaneously, the SR latch enters a “No Change” state, maintaining the input of whichever player pressed first; this effectively blocks the secondary input until the initial button is released, thereby preventing logic glitches and ensuring signal integrity.

5.1.4 Winner Indicator Circuit

Lane 1		Lane 2		Lane 3		Output of 3-input AND gate	Comparator		16-Segment Display	
Red Win	Blue Win	Red Win	Blue Win	Red Win	Blue Win		R>B	R<B	Red	Blue
0	0	X	X	X	X	0	X	X	-	-
X	X	0	0	X	X	0	X	X	-	-
X	X	X	X	0	0	0	X	X	-	-
0	0	0	0	X	X	0	X	X	-	-
X	X	0	0	0	0	0	X	X	-	-
0	0	X	X	0	0	0	X	X	-	-
0	1	0	1	1	0	1	0	1	-	W
0	1	1	0	0	1	1	0	1	-	W
1	0	0	1	0	1	1	0	1	-	W
0	1	1	0	1	0	1	1	0	W	-
1	0	0	1	1	0	1	1	0	W	-
1	0	1	0	0	1	1	1	0	W	-
0	1	0	1	0	1	1	0	1	-	W
1	0	1	0	1	0	1	1	0	W	-

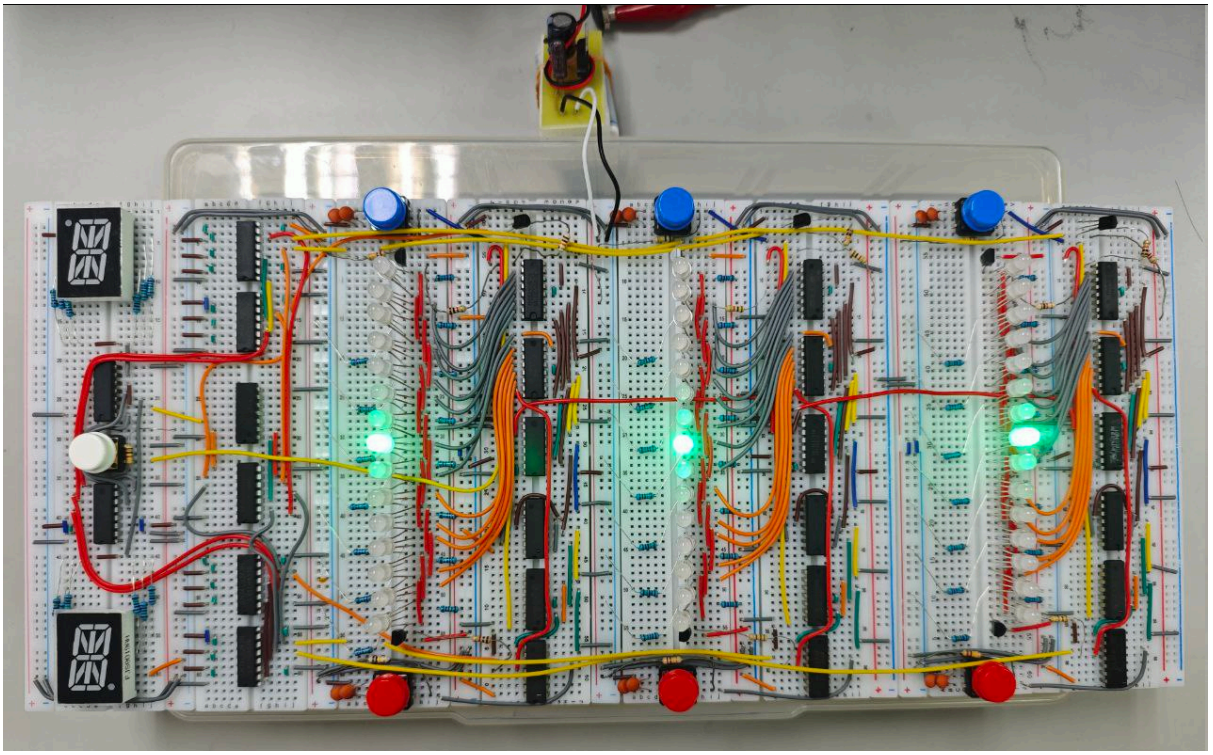
*Red Win means inverted $\overline{Y}_0$ from the first decoder of the lane, Blue Win means inverted $\overline{Y}_6$ from the second decoder of the lane.

Table 5.1.4

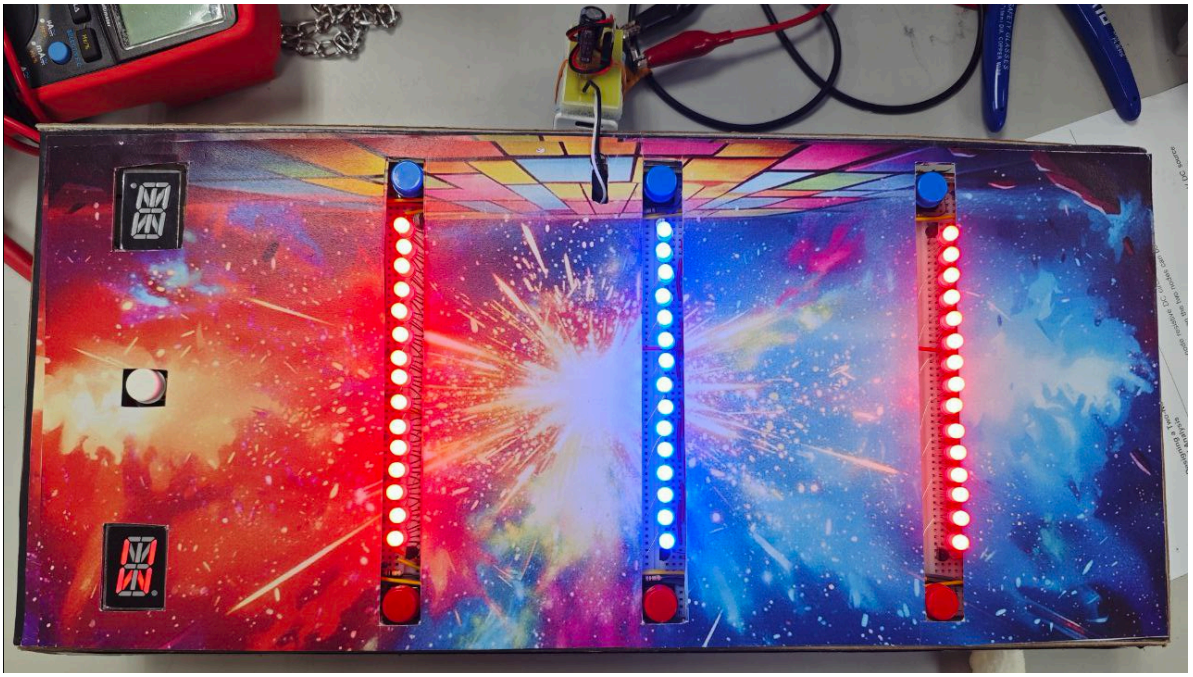
Table 5.1.4 shows the real-time behavior of the Winner Indicator Circuit. When the game starts, the output of comparator is blocked. Thus, the 16-segment displays won't light up even if there is a player with a higher score. Once all three lanes are ended, the W will show at the player's side with higher score to indicate the final winner of the game.

5.2 Final Prototype

Breadboard Circuit



Complete Prototype



5.3 Discussion

The implementation of the Photon Movement Control Circuit successfully achieved a smooth and responsive "Photon" movement across the 15 RGB LED lane. By utilizing the 74LS193 up/down counter, the circuit effectively tracks the Photon's position based on Player Blue (UP) and Player Red (DOWN) inputs. The cascading of two 74LS138 decoders to form a 4-to-16 decoder was executed successfully by using the counter's MSB (Q_D) to toggle the enable pins, allowing the circuit to drive the full range of LEDs from positions 0 to 14. The 74LS08 AND gate inhibitor, which monitored the boundary outputs ($\overline{Y}_0$ and $\overline{Y}_{14}$) successfully blocked the button input pulses once a player reached their respective goal, preventing the counter from wrapping around and ensuring the lane remained in a winning state.

The LED Array Circuit demonstrated high reliability in visualizing the game state. By using common anode RGB LEDs and allowing the three colors to share a single 150Ω resistor per LED, the circuit minimized component count while maintaining consistent brightness, as only one color is active at any time. The addition of PNP BJTs to drive all the red and blue cathodes from the 15 RGB LED ensured there is no overloading issue because the 74LS138 decoder lacks the necessary current sinking capacity for large amounts of LEDs. The middle green LEDs, driven directly by the decoders, provided a clear tracking of the Photon, ensuring players know well regarding the location of the Photon.

The Debounced Circuit and Glitch Prevention Circuit worked to ensure flawless input processing. The RC low-pass filters combined with 74LS14 Schmitt Triggers successfully eliminated mechanical contact bounce, providing a clean digital pulse for every button press. The 74LS00 SR latch solves the simultaneous press conflicts by maintaining a mutually exclusive logic state, ensuring that one clock signal (UP or DOWN) could transition only when another pin was held HIGH, preventing erratic jumps or logic glitches during fast-paced gameplay.

Finally, the Score Counter and Winner Indicator Circuit successfully automated the endgame process for all three lanes. The 74LS283 Adder accurately summed the winning signals from each lane, then correctly interpreted by the 74LS85 Comparator, which determined the overall winner based on the higher score. The use of 74LS86 XOR gates followed by the cascaded 3-input AND gates ensured that the final 16-segment display only showed the "W" indicator once the entire match was finished, providing a conclusive end to the game.

6.0 Project Management and Finance

6.1 Work Contribution

6.1.1 Circuit Design and Simulation

Task	Member
Photon Movement Control Circuit	TANG WEI PENG
LED Array Circuit	TANG WEI PENG WAN AMIERUL MUIZZUDDIN BIN WAN MAIZURIN
Debounced Circuit	TANG WEI PENG VISIRA KAARTIKA A/P MANOGRAN
Glitch Prevention Circuit	TANG WEI PENG UNG YI XIN
Winner Indicator Circuit	TANG WEI PENG

6.1.2 Report Writing

Section	Member
1.0 Theme and Title	TANG WEI PENG
2.0 Aim and Objectives	TANG WEI PENG VISIRA KAARTIKA A/P MANOGRAN
3.0 Introduction	TANG WEI PENG VISIRA KAARTIKA A/P MANOGRAN
4.0 Methodology	TANG WEI PENG UNG YI XIN
5.0 Result and Discussion	TANG WEI PENG
6.0 Project Management and Finance	UNG YI XIN WAN AMIERUL MUIZZUDDIN BIN WAN MAIZURIN
7.0 Conclusion	UNG YI XIN
8.0 Reference	All
9.0 Appendix	All

6.2 Component and Material List

Component / Material	Quantity	Price (RM)	Total Price (RM)
16 Segment Display	2	1.60	3.20
IC 7400	3	1.20	3.60
IC 7404	3	1.50	4.50
IC 7408	4	1.20	4.80
IC 7414	1	1.30	1.30
IC 7485	1	2.50	2.50
IC 7486	1	2.70	2.70
IC 74138	6	2.10	12.60
IC 74193	3	3.40	10.20
IC 74283	4	3.50	14.00
RGB LED	45	0.60	27.00
Resistor 150 Ω	57	0.02	1.14
Resistor 1 kΩ	9	0.02	0.18
Resistor 100 kΩ	6	0.02	0.12
4 pins push button	6	0.60	3.60
Capacitor 0.1μF	6	0.03	0.18
PNP 2N2907 Transistor	6	0.40	2.40
Total Price (RM)			94.02

6.3 Ethical Consideration

The design and implementation of the Photon War system were performed by putting the principles of the engineering profession into considerations and it is suitable for the corresponding professional standards inclusive of the IEEE and NSPE Codes of Ethics. Ethical responsibility throughout the system lifecycle was emphasized on such issues as safety, reliability, equity, sustainability, and professional integrity.

From a safety point of view, the whole system works within only 5 V which is a low voltage that greatly minimizes the chances of electrical hazards. All the circuits of LEDs had current-limiting resistors of 150 Ω to regulate current through them and avoid heating up. All wiring connections and the PNP transistor stages were checked to safely handle the currents of LEDs within the specification for operation. These are now tested under various operational conditions to identify and mitigate against short circuits, excessive current draws, thermal stress, and risk. This makes the entire operation safe and assures users, including children and seniors, of safety during game operation.

To guarantee the reliability and proper functioning of the system, the standard digital design practices were included. The switching inputs were debounced using RC debounce circuits and Schmitt trigger inverters to avoid spurious activation due to switch bounce. Glitch prevention logic was used to handle the simultaneous activation of the inputs to guarantee the determinism of the sequential logic.

The fairness and impartiality were ensured through symmetric implementation at both hardware and logical levels. This was achieved through identical hardware and logical circuits for both players. Similar hardware and logical structures were implemented to ensure an identical system response without any internal bias. The game results and scoring were decided purely at the objective combinational logical level.

Sustainability in terms of resource use has also been taken into account. The use of components has been maximized by ensuring that the resistor is safely sharable by the RGB LED components. Low-cost components have been considered so that the project cost is only at RM 94.02. This will prevent the use of extra components that end up in landfills. The circuit has been designed in a way that it is reusable.

Finally, compliance with professional and academic standards was strictly maintained. All technical content, circuit designs, and experimental results were accurately reported without fabrication or misleading claims. Facilitators and relevant references were properly credited, and the project was designed to demonstrate digital electronics theories clearly and truthfully within a sci-fi themed game context, in alignment with IEEE and NSPE ethical guidelines.

In conclusion, ethical considerations were integrated into this project through adherence to safety standards, reliable and fair system design, sustainable engineering practices, and professional integrity, reflecting the ethical obligations expected of practicing engineers.

6.4 Future Recommendations

In this project, there are some modifications that can be done to enhance the circuit features. One of the adjustments that could be done in future is adding more lanes for Photon War. This could increase the difficulty of the game and enhances the enjoyment of the game. By adding more lanes, players have to monitor multiple Photon movements simultaneously. Thus, this modification demands faster reaction time and better coordination. This could be done by duplicating the existing lane circuitry without altering its fundamental operating principles.

From another perspective, the modification that can be done is to incorporate a lane selection circuit to activate a specific chosen lane while disabling the others. This allows the players to move a cursor via the “Left” and “Right” selection button to select a lane during Photon War, effectively preventing players from pressing multiple buttons across different lanes simultaneously. This feature would provide a higher degree of fairness and reduce the chance of cheating, bringing a better and more balanced gaming experience to the users.

7.0 Conclusion

In conclusion, the Photon War mini project was successfully designed, implemented and tested in accordance with the objectives of this mini project. Practical application of digital logic concepts are demonstrated through the integration of counters, decoders, combinational logic and sequential circuits to form an interactive sci-fi themed game. When a player presses their assigned button, the signal is processed through the 74LS193 up/down counter and cascaded 74LS138 decoders to shift the green Photon LED incrementally toward the opponent's side. The system features a reset button that utilizes the counter's preset feature to return all Photons to their center starting positions, ensuring a fair start for every match. Furthermore, the circuit accurately determines an overall winner by using 74LS283 adders and a 74LS85 comparator to tally conquered lanes and display a "W" on the winner's 16-segment display once the game concludes.

From this project, we are able to master all the knowledge gained from all the previous laboratory sessions and apply them in designing advanced circuits, such as RC debounce filters, Schmitt Trigger conditioning, and SR latch signal arbitration for glitch prevention. Although there were some problems faced when carrying out this project, such as resolving mechanical button bounce and preventing logic errors when buttons are pressed simultaneously, but thanks to the cooperation of all team members and also guidance from lecturers, we were able to troubleshoot and construct the circuit successfully.

8.0 References

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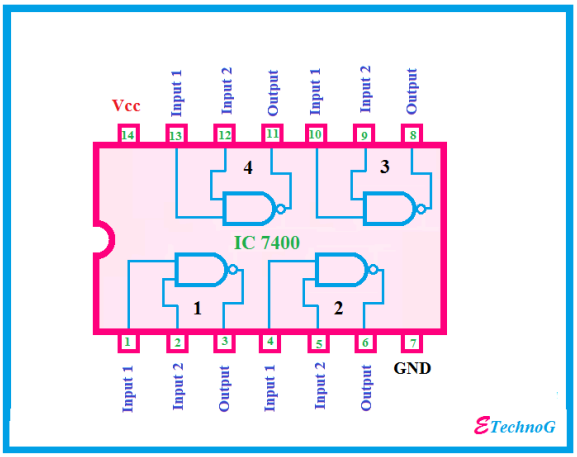
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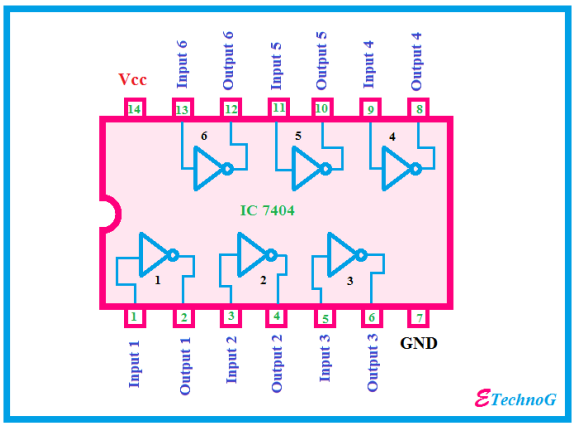
9.0 Appendices

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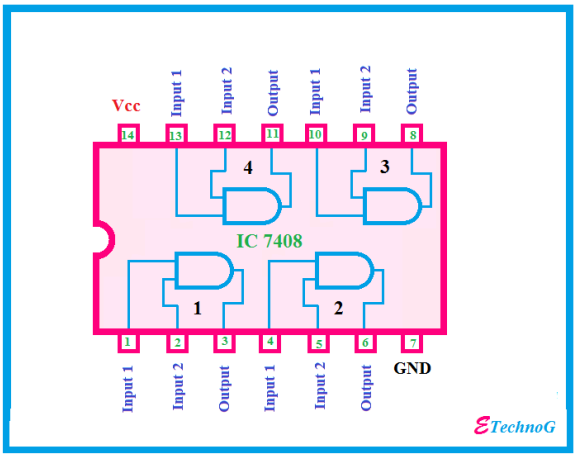
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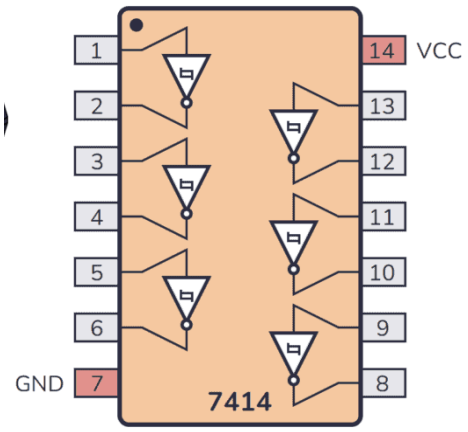
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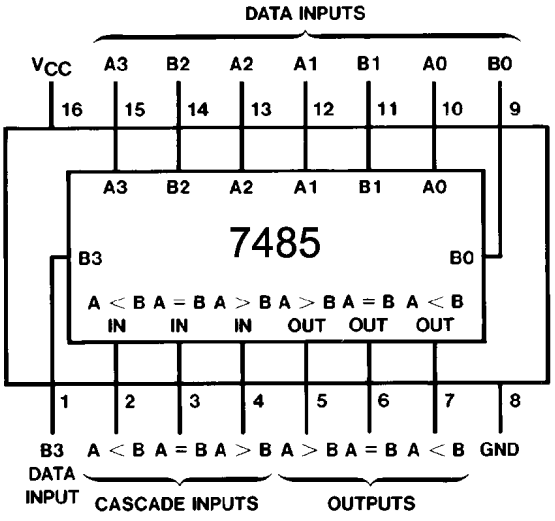
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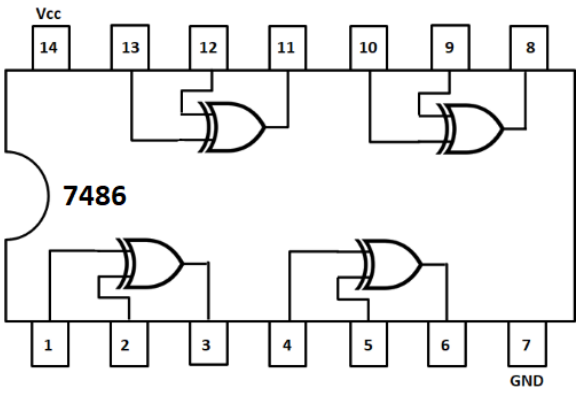
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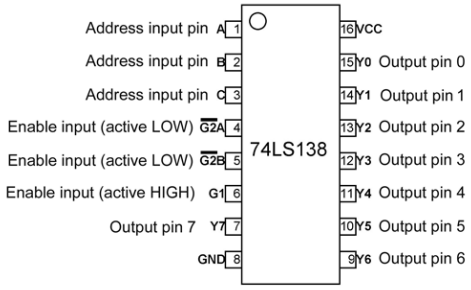
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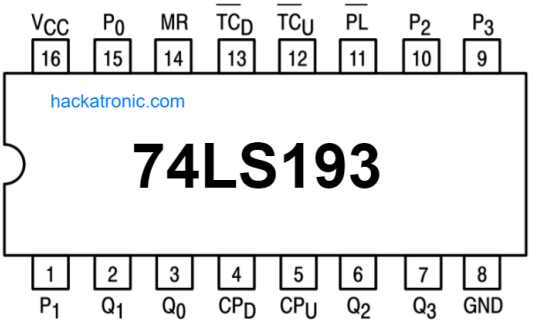
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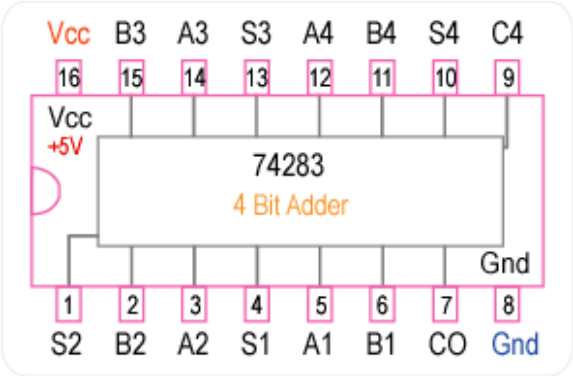
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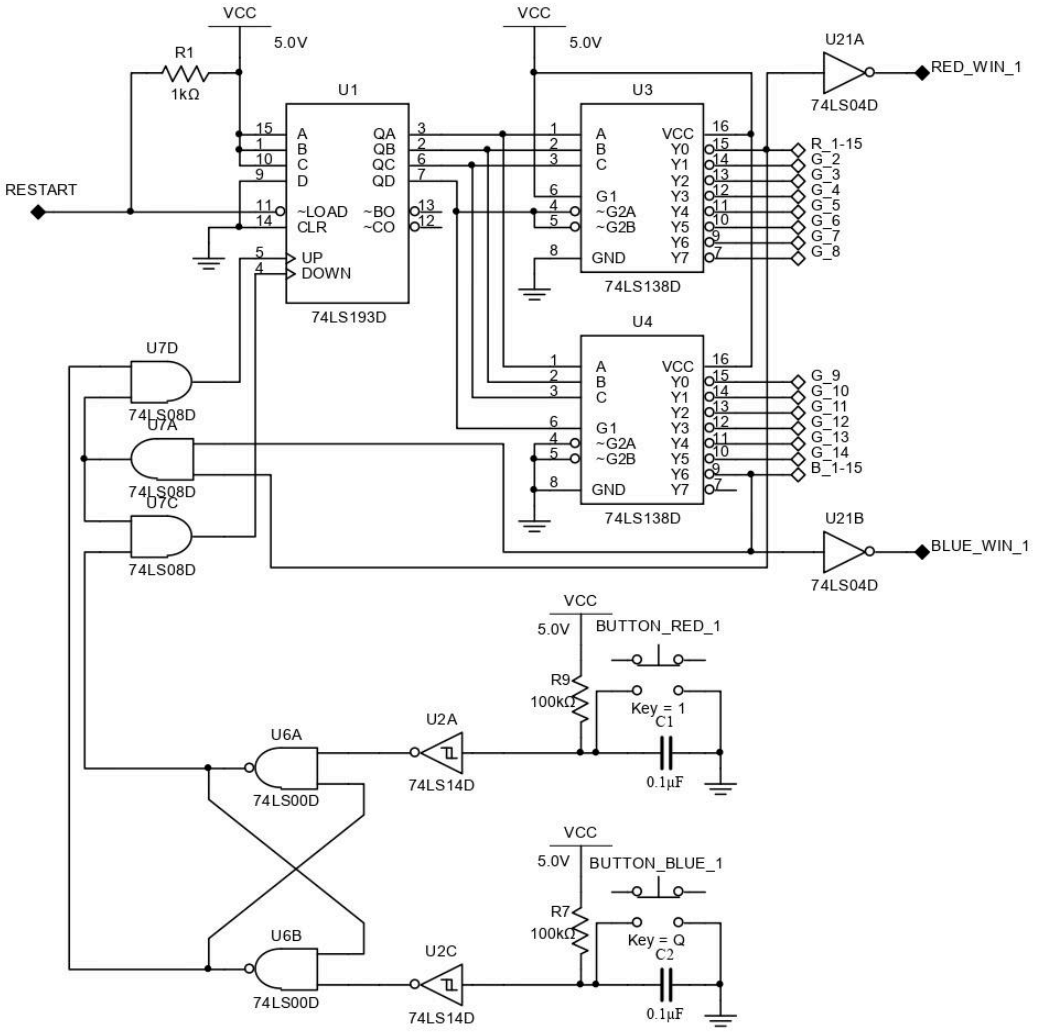
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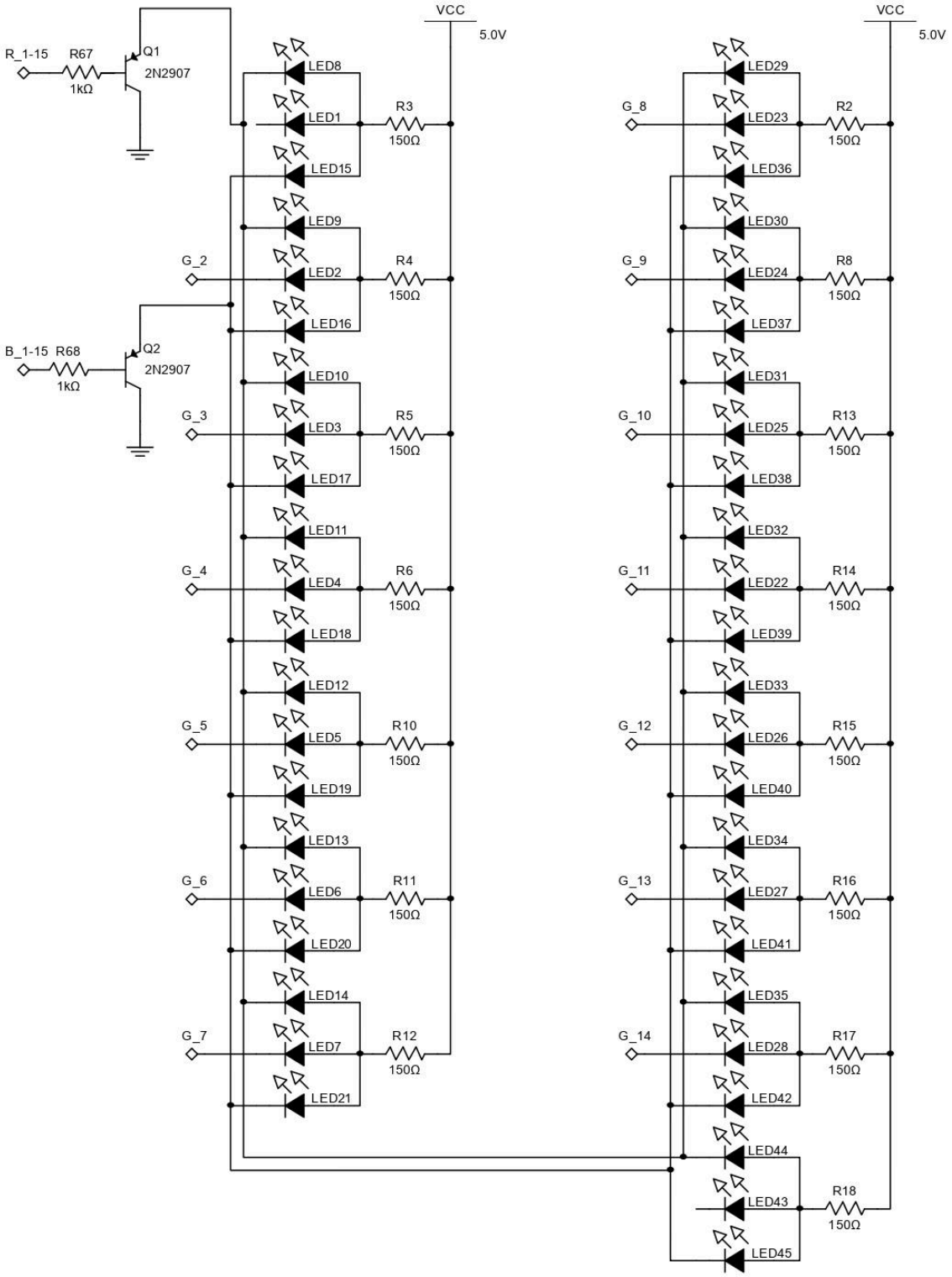


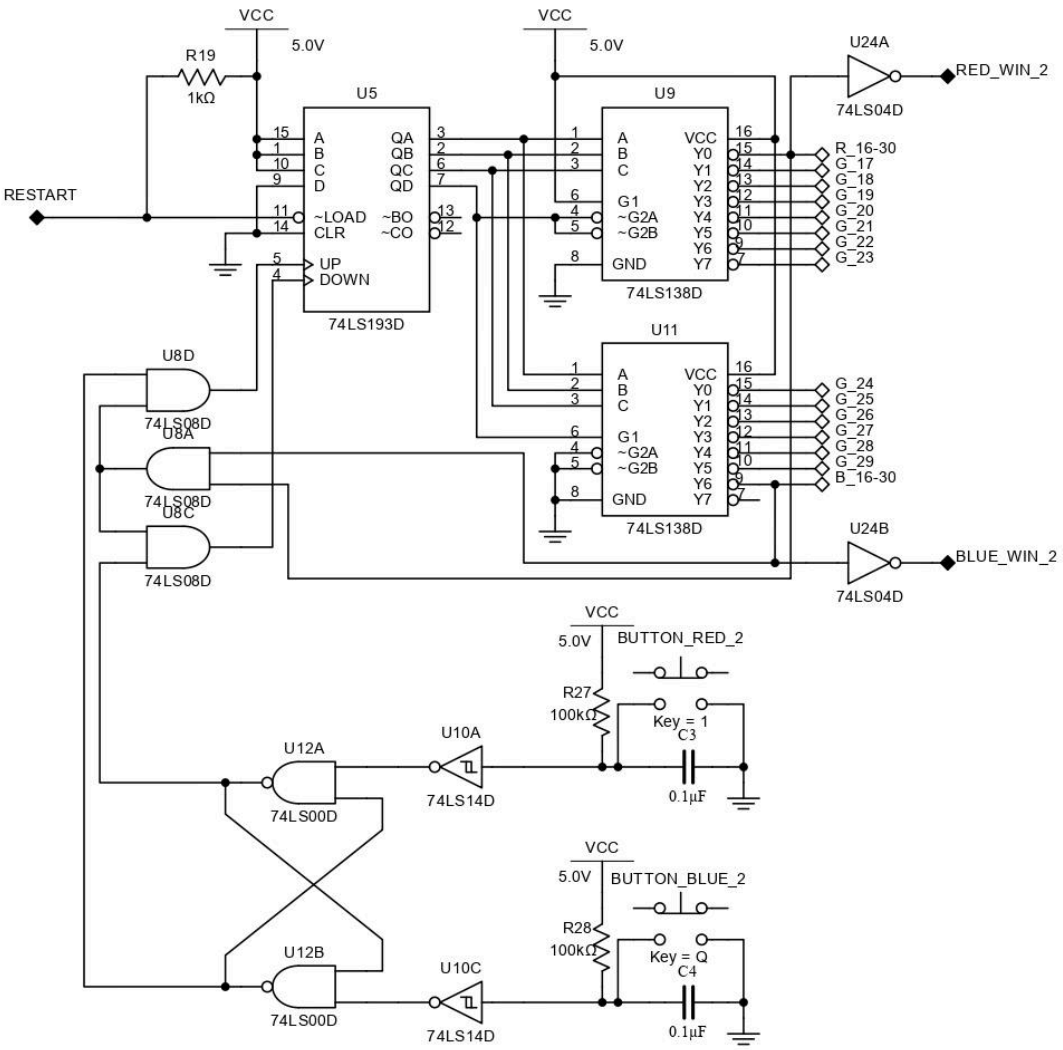
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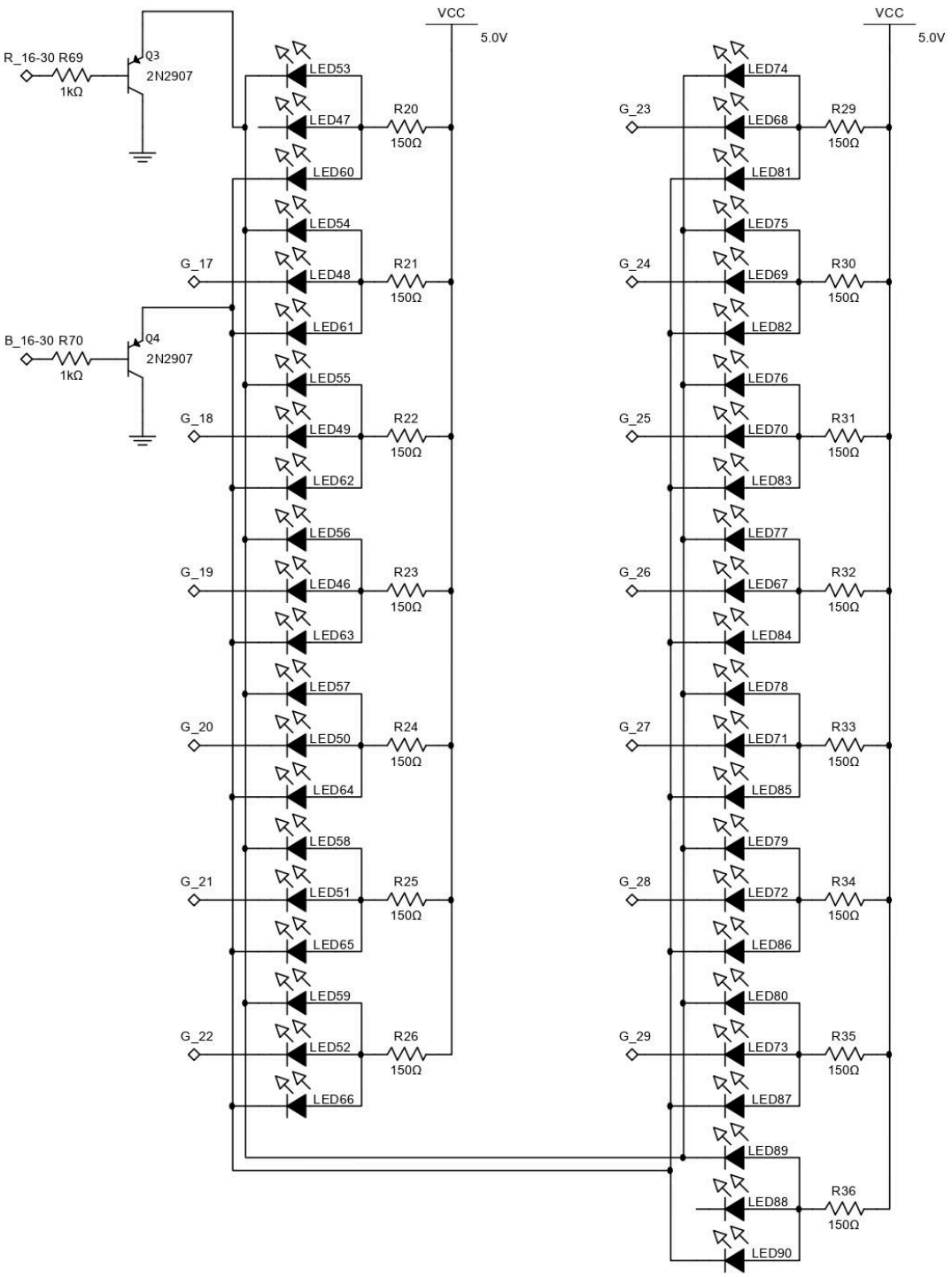


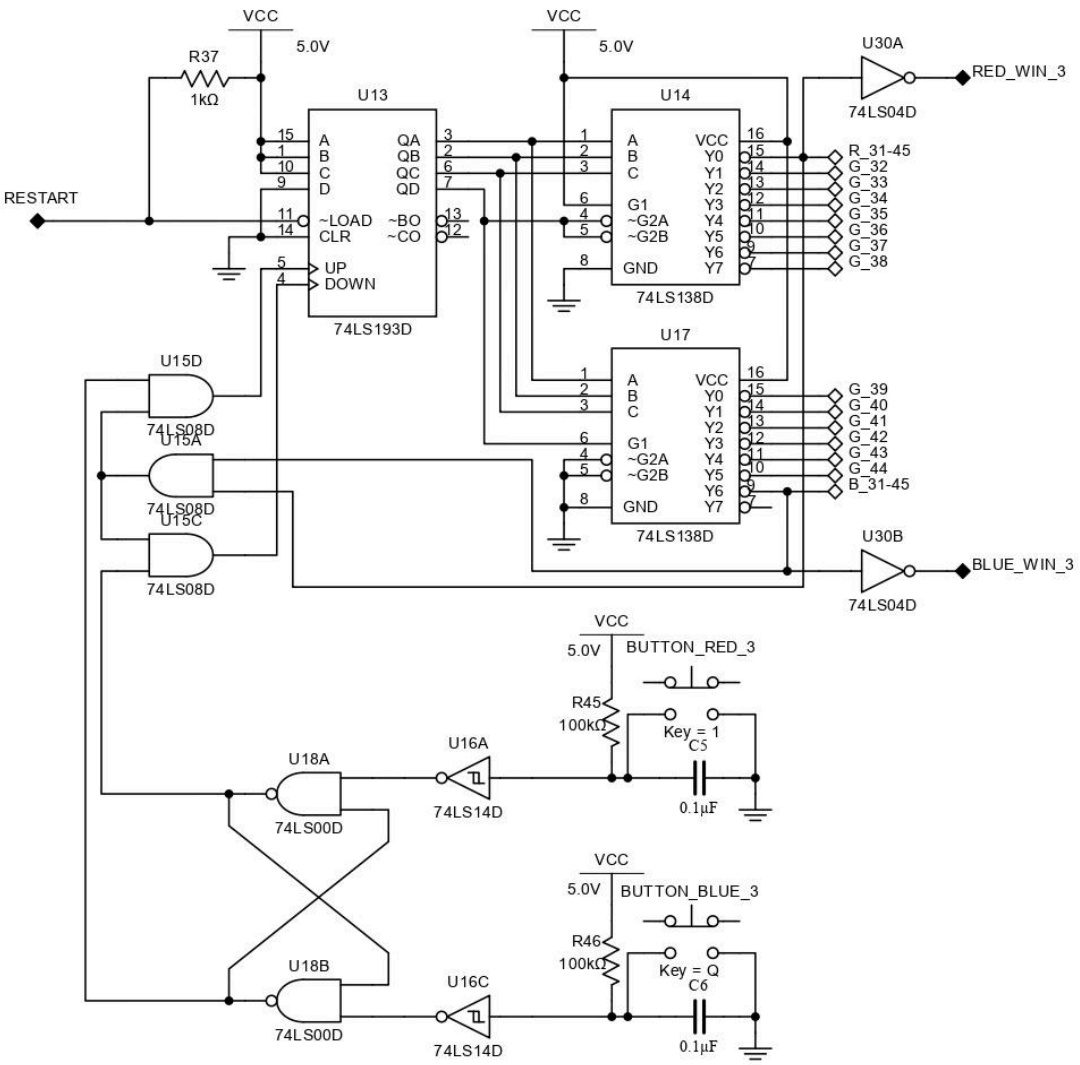
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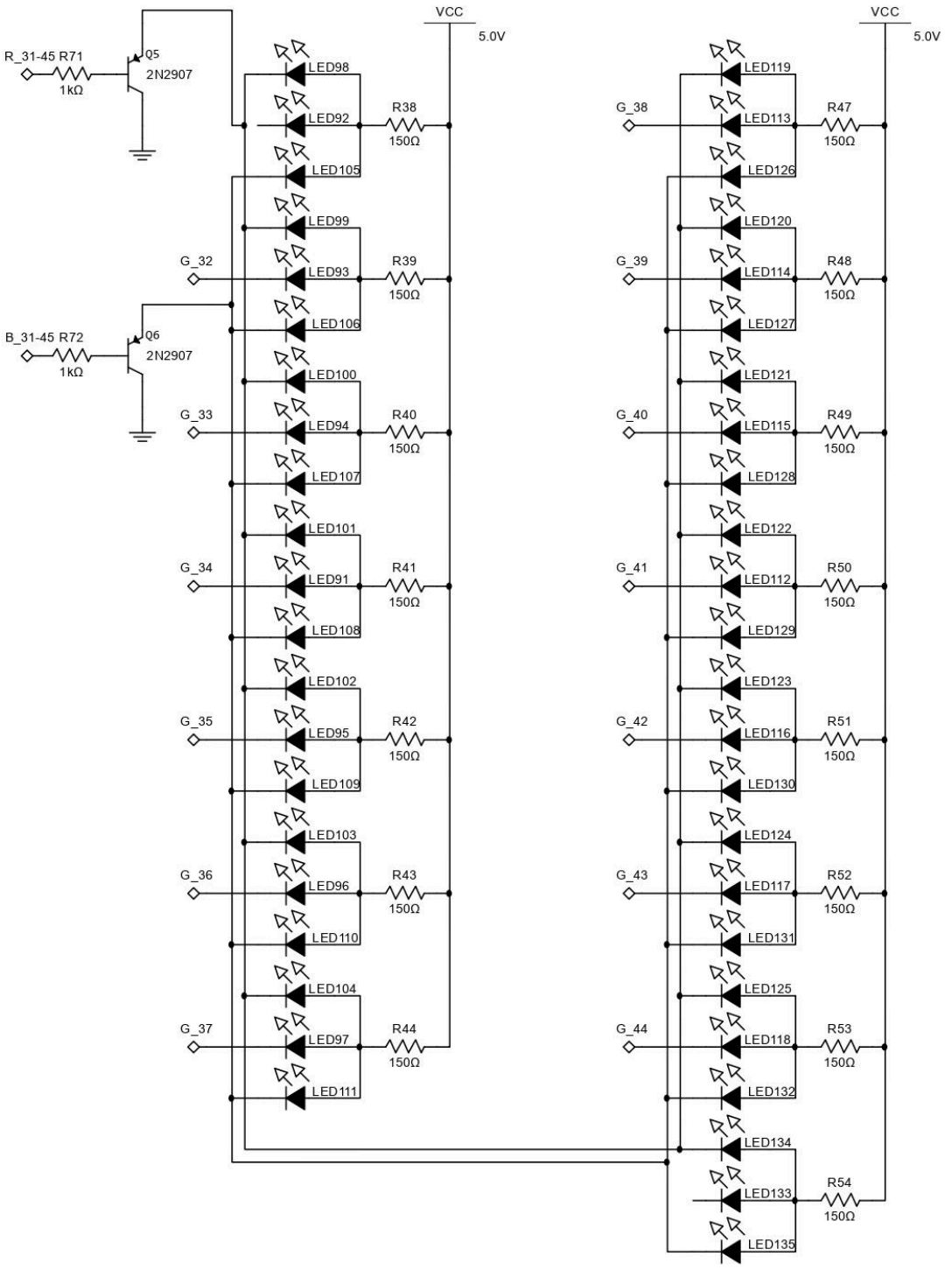


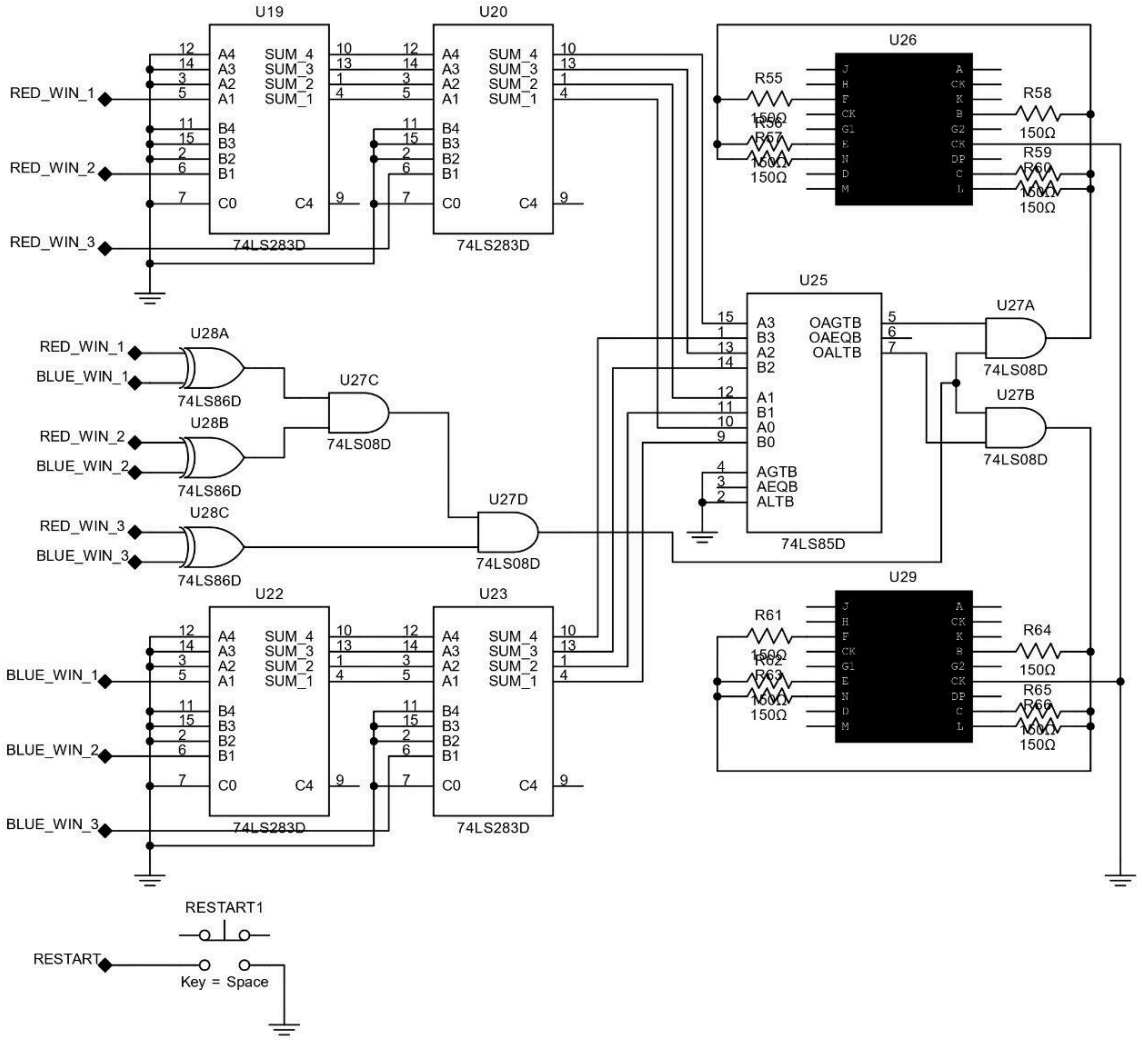

VCC
5.0V
R_1-15
R67
1kΩ
Q1
2N2907
B_1-15
R68
1kΩ
Q2
2N2907
LED8
LED1
LED15
LED9
LED2
LED16
LED10
LED3
LED17
LED11
LED4
LED18
LED12
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LED19
LED13
LED6
LED20
LED14
LED7
LED21
R3
R4
R5
R6
R10
R11
R12
150Ω
G_2
G_3
G_4
G_5
G_6
G_7
LED29
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R8
R13
R14
R15
R16
R17
R18
G_8
G_9
G_10
G_11
G_12
G_13
G_14


VCC
5.0V
R19
1kΩ
U5
RESTART
~LOAD
CLR
UP
DOWN
~BO
~CO
74LS193D
VCC
5.0V
U9
A
B
C
G1
~G2A
~G2B
GND
VCC
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
74LS138D
U24A
74LS04D
RED_WIN_2
R_16-30
G_17
G_18
G_19
G_20
G_21
G_22
G_23
U11
74LS138D
G_24
G_25
G_26
G_27
G_28
G_29
B_16-30
U24B
74LS04D
BLUE_WIN_2
U8D
74LS08D
U8A
74LS08D
U8C
74LS08D
VCC
5.0V
BUTTON_RED_2
R27
100kΩ
Key = 1
C3
0.1µF
U12A
74LS00D
U10A
74LS14D
VCC
5.0V
BUTTON_BLUE_2
R28
100kΩ
Key = Q
C4
0.1µF
U12B
74LS00D
U10C
74LS14D


VCC
5.0V
R_16-30 R69
1kΩ
Q3
2N2907
B_16-30 R70
1kΩ
Q4
2N2907
G_17
G_18
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R32
R33
R34
R35
R36
150Ω


VCC
5.0V
R37
1kΩ
U13
74LS193D
RESTART
~LOAD
CLR
~BO
~CO
UP
DOWN
U14
74LS138D
U17
74LS138D
GND
G1
~G2A
~G2B
U30A
74LS04D
RED_WIN_3
R_31-45
G_32
G_33
G_34
G_35
G_36
G_37
G_38
G_39
G_40
G_41
G_42
G_43
G_44
B_31-45
U30B
74LS04D
BLUE_WIN_3
U15D
74LS08D
U15A
74LS08D
U15C
74LS08D
BUTTON_RED_3
R45
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Key = 1
C5
0.1µF
U16A
74LS14D
U18A
74LS00D
BUTTON_BLUE_3
R46
100kΩ
Key = Q
C6
0.1µF
U16C
74LS14D
U18B
74LS00D


VCC
5.0V
R_31-45 R71
1kΩ
Q5
2N2907
B_31-45 R72
1kΩ
Q6
2N2907
G_32
G_33
G_34
G_35
G_36
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G_38
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G_40
G_41
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R39
R40
R41
R42
R43
R44
R47
R48
R49
R50
R51
R52
R53
R54
150Ω


U19
U20
U26
RED_WIN_1
RED_WIN_2
RED_WIN_3
A4
A3
A2
A1
B4
B3
B2
B1
C0
C4
SUM_4
SUM_3
SUM_2
SUM_1
74LS283D
R55
R56
R57
R58
R59
R60
150Ω
U25
A3
B3
A2
B2
A1
B1
A0
B0
AGTB
AEQB
ALTB
OAGTB
OAEQB
OALTB
74LS85D
U27A
U27B
U27C
U27D
74LS08D
U28A
U28B
U28C
BLUE_WIN_1
BLUE_WIN_2
BLUE_WIN_3
74LS86D
U22
U23
U29
R61
R62
R63
R64
R65
R66
RESTART1
RESTART
Key = Space